MILITARY SPECIFICATION

TEST PLUG ASSEMBLY

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of test plug assembly used on XM455 projectile.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-C-48019	-Connector, electrical
MIL-P-116	-Preservation, Methods of
MIL-S-45743	-Soldering, Manual Type, High Reliability Electrical, Electronic, Instruments, Communication, and Radar for Aerospace, and Control Systems, Procedure for

STANDARDS

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168	-Lot Numbering of Ammunition
MIL-STD-1169	-Packaging, Packing and Marking for Shipments of Inert Ammunition Components

FSC 1190

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MS11625-4181	-Ring Retaining
MS16995-27	-Screw, Cap, Soc Hd
MS16997-97	-Screw, Cap, Socket Head
MS16998-32	-Cap Screw
MS35270-59	-Machine Screw
MS51021-9	-Set Screw

DRAWINGS

ORDNANCE CORPS

8808362	-Packing and Marking for Storage and Shipments
9220308	-Connector, Electrical
10522826	-Connector Assembly
10522828	-Modified Cap Screw
10522829	-Screw, Cap, Socket Head
10522830	-Gasket
10522833	-Dust, Cover
10522834	-Gasket, Bolt
10522835	-Body (Trainer)
10523295	-Socket, Connector
10533952	-Set Screw
10533960	-Screw, Machine
10533972	-Test Plug Assembly (Trainer)
10534293	-Cap (Trainer)
ET-10533972	-Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The test plug assembly shall comply with all requirements specified on the drawing (dwg) 10533972 with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

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3.3 Circuit resistance. (See dwg 10533972). The circuit resistance shall comply with table I when tested as specified in 4.4.1.

TABLE I

Circuit resistance (see dwg 10533972)

From Connector Assembly	To Socket connector	Ohms Max.
A	9	
C	7	

3.4 Insulation resistance. (see dwg 10533972). The insulation resistance of any circuit shall not be less than 100 megohms when subjected to a potential of 550±50 volts direct current (vdc) with the current limited to 5 milliamperes (ma) when tested as specified in 4.4.2.

3.5 Connector assembly (see dwg 10523295). Connector assembly dwg. 9220308 is the same as connector assembly dwg. 10523295. The connector assembly shall comply with MIL-C-48019(MU).

3.6 Workmanship.

3.6.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristics.

3.6.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.3 Burrs. Parts shall be free from burrs.

3.6.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.6.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.6.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Tabel III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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TABLE II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

TABLE III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Screw, Cap, Socket head (see dwg 10522829) Covering a detail of dwg 10533972.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101 Objective evidence (see 6.2 for machine screw in accordance with MS35270-59	Visual
	102 Hole present	Visual
Minor:		
	201 Workmanship (see 3.6)	Visual/Manua

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4.3.2.1.2 Screw, cap, soc hd (see MS16995-27); set screw (see dwg 10533952); gasket, bolt (see dwg 10522834) ring, retaining (see MS16625-4181); set screw (see 51021-9) covering details of dwg 10533972.

Categories	Characteristics	Method of Inspection

Critical: None defined

Major:
101. Objective evidence (see 6.2) for component in accordance with applicable drawing or standard....Visual

Minor:
201. Workmanship(see 3.6).....Visual/Manual

4.3.2.1.3 Gasket, dust cover (see dwg 10534657) covering a detail of dwg 10533972.

Categories	Characteristics	Method of Inspection

Critical: None defined

Major:
101. Material (see 3.1).....Visual
102. No tear, crack or gouge .....Visual

Minor:
201. Workmanship (see 3.6) .....Visual/Manual

4.3.2.1.4 Dust cover (see dwg 10522833) covering a detail of dwg 10533972.

Cateogries	Characteristics	Method of Inspection

Critical: None defined

Major:
101. Material (see 3.1).....Visual

Minor:
201. Workmanship (see 3.6).....Visual/Manual

4.3.2.1.5 Body (trainer) (see dwg 10522835) covering a detail of dwg 10533972.

Categories	Characteristics	Method of Inspection

Critical: None defined

Major:
101. Concentricity of right outside diameter.....Gage

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Categories	Characteristics	Method of Inspection
	102. Right outside diameter (3.789)	Gage
	103. Width of outside slot, min (1.54)	Gage
	104. Bottom outside diameter (3.764)	Gage
	105. Length of bottom outside diameter (1.110)	Gage
	106. Left inside diameter (1.52)	Gage
	107. Length of left inside diameter (.12)	Gage
	108. Pitch diameter of tapped hole in side of left cavity	Gage
	109. Minor diameter of tapped hole in side of left cavity	Gage
	110. Depth of left slot (.380)	Gage
	111. Right inside straight diameter (1.830)	Gage
	112. Total length (1.45)	Gage
	113. Material (see 3.1)	Visual
	114. Dye finish present, no bare spot	Visual
	115. Material (see 3.1)	Visual
Minor:		
	201. Radius present	Visual
	202. Surface finish	Visual
	203. Workmanship (see 3.6)	Visual/Manual

4.3.2.1.6 Gasket (see dwg 10522830) covering a detail of dwg 10533972.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major:		
	101. Outside diameter, min	*Measure
	102. Inside diameter, min	*Measure
	103. Thickness, min	*Measure
	104. No splits, cracks and properly formed	Visual
	105. Material (see 3.1)	Visual

*Measure with suitable scale.

Minor:		
	201. Workmanship (see 3.6)	Visual/Manual

4.3.2.1.7 Key (see dwg 10522823) covering a detail of dwg 10533972.

Categories	Characteristics	Method of Inspection
Critical: None defined		

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Major:
101 Width of top lugs, min (.251).....Gage
102 Width of key, min (1.520).....Gage
103 Counterbored thru hole present.....Visual
104 Material (see 3.1).....Visual

Minor:
201 Surface finish.....Visual
202 Protective finish present, no bare spot.....Visual
203 Workmanship (see 3.6).....Visual/Manual

4.3.2.1.8 Modified cap screw (see dwg 10522828) covering a detail of dwg 10533972.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major:		
	101 Length, max.	Gage
	102 Objective evidence (see 6.2) for screw in accordance with MS16998	Visual
Minor:		
	201 Protective finish present no bare spot	Visual
	202 Workmanship (see 3.6)	Visual/Manual

4.3.2.1.9 Cap (trainer)(see dwg 10534293) covering a detail of dwg 10533972.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major:		
	101 Pitch diameter of tapped hole left diameter.	Gage
	102 Minor diameter of tapped hole left diameter.	Gage
	103 Left diameter (1.50)	Gage
	104 Diameter of thru hole (.19)	Gage
	105 Location of thru hole	Gage
	106 Length of right diameter, min (.10)	Gage

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107 Location of tapped hole in left diameter....Gage
108 Total thickness, min.....Gage
109 Material (see 3.1).....Visual

Minor:
201 Surface finish.....Visual
202 Protective finish present no bare spot.....Visual
203 Workmanship (see 3.6).....Visual/Manual

4.3.2.1.10 Screw, cap, socket head (see dwg 10522829) covering a detail of dwg 10533972

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major:		
	101 Objective evidence (see 6.2) for screw in accordance with MS16997	Visual
Minor:		
	201 Protective finish present no bare spot	Visual
	202 Workmanship (see 3.6)	Visual/Manual

4.3.2.1.11 Connector assembly (see dwg 10522826) covering a detail of dwg 10533972.

Categories	Characteristics	Method of Inspection
Critical: None defined		
Major:		
	101 Soldering in accordance with MIL-S-45743	Visual/Manual
	102 Material (see 3.1)	Visual
	103 Proper assembly of sockets	Visual
Minor:		
	201 Workmanship (see 3.6)	Visual/Manual

4.3.2.1.12 Test plug assembly (trainer)(see dwg 10533972 before potting

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Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101 Material (see 3.1)	Visual
	102 Sealant on connector assembly present	Visual
	103 Machine screw 10522829 rotates freely	Manual
	104 Dust cover tight	Manual
	105 Correct number of gaskets 10522834	Visual
	106 All parts correctly assembled, no part missing	Visual
Minor:		
	201 Workmanship (see 3.6)	Visual/Manual

4.3.2.1.13 Test plug assembly (trainer)(see dwg 10533972.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101 Material (see 3.1)	Visual
	102 Proper potting	Visual
Minor:		
	201 Workmanship (see 3.6)	Visual/Manual

4.3.2.1.14 Preparation for delivery (see 5.1)

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101 Preservation, packaging, packing and marking in accordance with MIL-STD-1169	Visual/Manual
Minor:	None defined	

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4.3.2.1.15 Bag, packing for repair parts (see dwg 8883539 covering dwg part number 8883539-8) and 5.2.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101 Material (see 3.1)	Visual
	102 Length before packing, min	*Measure
	103 Width before packing, min	*Measure
	104 Properly packed	Visual/Manual
	105 Proper heat seal	Visual/Manual
	106 Objective evidence (see 6.2) for heat seal test in accordance with MIL-P-116	Visual/Manual
	107 No tear or hole	Visual
	*Measure with suitable scale	
Minor:		
	201 Marking	Visual
	202 Workmanship (see 3.6)	Visual/Manual

4.3.2.1.16 Packing and marking for storage and shipments (see dwg 8808362) and 5.2.

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101 Gross load weights	*Measure
	102 Contents tight in package	Manual
	103 Correctly packed	Visual/Manual
	104 Material (see 3.1)	Visual
	*Determine weight with suitable scale	
Minor:		
	201 Marking	Visual
	202 Workmanship (see 3.6)	Visual/Manual

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4.3.3 Testing. All tests shall be performed. Tests need not be performed in the order specified.

Test	Test Classification
a. Circuit Resistance	Major
b. Insulation Resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Circuit resistance (see 3.3). Each test plug shall be tested in accordance with 4.4.1. If any item fails to comply with 3.3, the item shall be rejected and removed from the lot.

4.3.3.2 Insulation resistance (see 3.4). Each test plug shall be tested in accordance with 4.4.2. If any item fails to comply with 3.4, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-10533972. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for monor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Circuit resistance (see 3.3). The circuit resistance shall be measured in accordance with Table I (see 3.3) with approved measuring equipment.

4.4.2 Insulation resistance (see 3.4). The insulation resistance shall be measured with approved insulation resistance measuring equipment in accordance with Table IV.

TABLE IV
Insulation resistance (see dwg 10533972)

Positive Potential to connector assembly pin	Socket connector Negative potential to all other pins except
A	9
C	7

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The test plug shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The test plug shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The test plug shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. The test plug shall be packaged in accordance with dwg 8883539 and 8808362 using the specified moisture barrier.

5.2.1.2 Level C. The test plug shall be packaged in accordance with dwg 8808362 except that the moisture barrier may be omitted.

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5.2.2 Packing.

5.2.2.1 Level A and C. The test plug shall be packed in accordance with dwg 8883539 and 8808362.

5.2.3 Marking. Marking shall be as specified in accordance with dwg 8883539 and 8808362 as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Production or Repair Parts.
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 or specification when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN; SMUPA-QA-I.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1190-A061

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DEPARTMENT OF THE ARMY

POSTAGE AND FEES PAID
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OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-T-48104(MU) Test Plug Assembly

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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